

cross-pollination of two plants, Mendel removed the immature stamens of a plant before they produced pollen and then dusted pollen from another plant onto the altered flowers (see Figure 14.2). Each resulting zygote then developed into a plant embryo encased in a seed (a pea). His method allowed Mendel to always be sure of the parentage of new seeds.

Mendel chose to track only those characters that occurred in two distinct, alternative forms, such as purple or white flower color. He also made sure that he started his experiments with varieties that were **true-breeding**—that is, over many generations of self-pollination, these plants had produced only the same variety as the parent plant. For example, a plant with purple flowers is true-breeding if the seeds produced by self-pollination in successive generations all give rise to plants that also have purple flowers.

In a typical breeding experiment, Mendel cross-pollinated two contrasting, true-breeding pea varieties—for example, purple-flowered plants and white-flowered plants (see Figure 14.2). This mating, or *crossing*, of two true-breeding varieties is called **hybridization**. The true-breeding parents are referred to as the **P generation** (parental generation), and their hybrid offspring are the **F₁ generation** (first filial generation, the word *filial* from the Latin word for “son”). Allowing these F₁ hybrids to self-pollinate (or to cross-pollinate with other F₁ hybrids) produces an **F₂ generation** (second filial generation). Mendel usually followed traits for at least the P, F₁, and F₂ generations. Had he stopped his experiments with the F₁ generation, the basic patterns of inheritance would have eluded him. Mendel’s quantitative analysis of the F₂ plants from thousands of genetic crosses like these allowed him to deduce two fundamental principles of heredity, now called the law of segregation and the law of independent assortment.

The Law of Segregation

The explanation of heredity most widely in favor during the 1800s was the “blending” hypothesis, the idea that genetic material contributed by the two parents mixes just as blue and yellow paints blend to make green. This hypothesis predicts that over many generations, a freely mating population will give rise to a uniform population of individuals, something we don’t see.

If the blending model were correct, the F₁ hybrids from Mendel’s cross between purple-flowered and white-flowered pea plants would have been pale purple flowers, a trait intermediate between those of the P generation. Notice in Figure 14.2 that the experiment produced a very different result: All the F₁ offspring had flowers of the same color as the purple-flowered parents. What happened to the white-flowered plants’ genetic contribution to the hybrids? If it were lost, then the F₁ plants could produce only purple-flowered offspring in the F₂ generation. But when Mendel allowed the F₁ plants to self- or cross-pollinate and

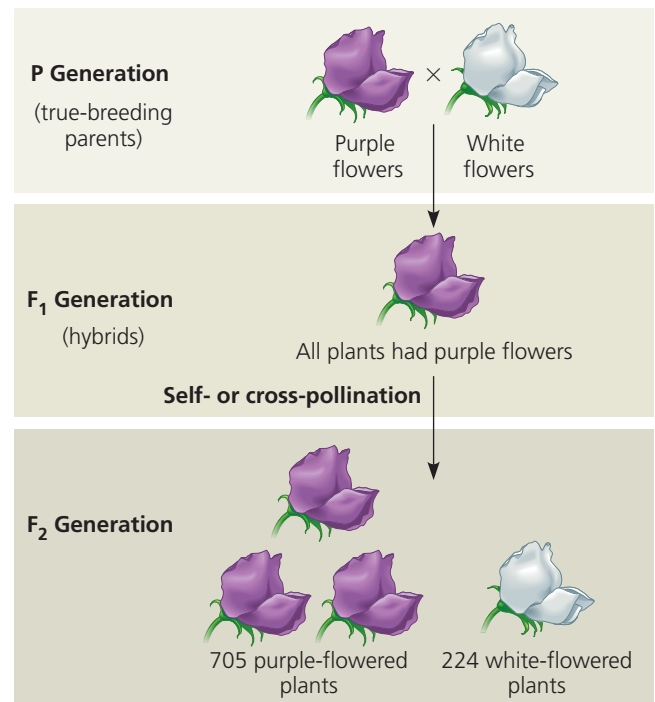
planted their seeds, the white-flower trait reappeared in the F₂ generation. The blending hypothesis is also inconsistent with this reappearance of traits after they’ve skipped a generation.

Mendel used very large sample sizes and kept accurate records of his results: 705 of the F₂ plants had purple flowers, and 224 had white flowers. These data fit a ratio of approximately three purple to one white (**Figure 14.3**). Mendel reasoned that the heritable factor for white flowers did not disappear in the F₁ plants but was somehow hidden, or masked,

▼ Figure 14.3 Inquiry

When F₁ hybrid pea plants self- or cross-pollinate, which traits appear in the F₂ generation?

Experiment Mendel crossed true-breeding purple-flowered plants and white-flowered plants (crosses are symbolized by ×). The resulting F₁ hybrids were allowed to self-pollinate or were cross-pollinated with other F₁ hybrids. The F₂ generation plants were then observed for flower color.



Results Both purple-flowered and white-flowered plants appeared in the F₂ generation, in a ratio of approximately 3:1.

Conclusion The “heritable factor” for the recessive trait (white flowers) had not been destroyed, deleted, or “blended” in the F₁ generation but was merely masked by the presence of the factor for purple flowers, which is the dominant trait.

Data from G. Mendel, Experiments in plant hybridization, *Proceedings of the Natural History Society of Brunn* 4:3–47 (1866).

WHAT IF? If you mated two purple-flowered plants from the P generation, what ratio of traits would you expect to observe in the offspring? Explain. What might Mendel have concluded if he stopped his experiment after the F₁ generation?